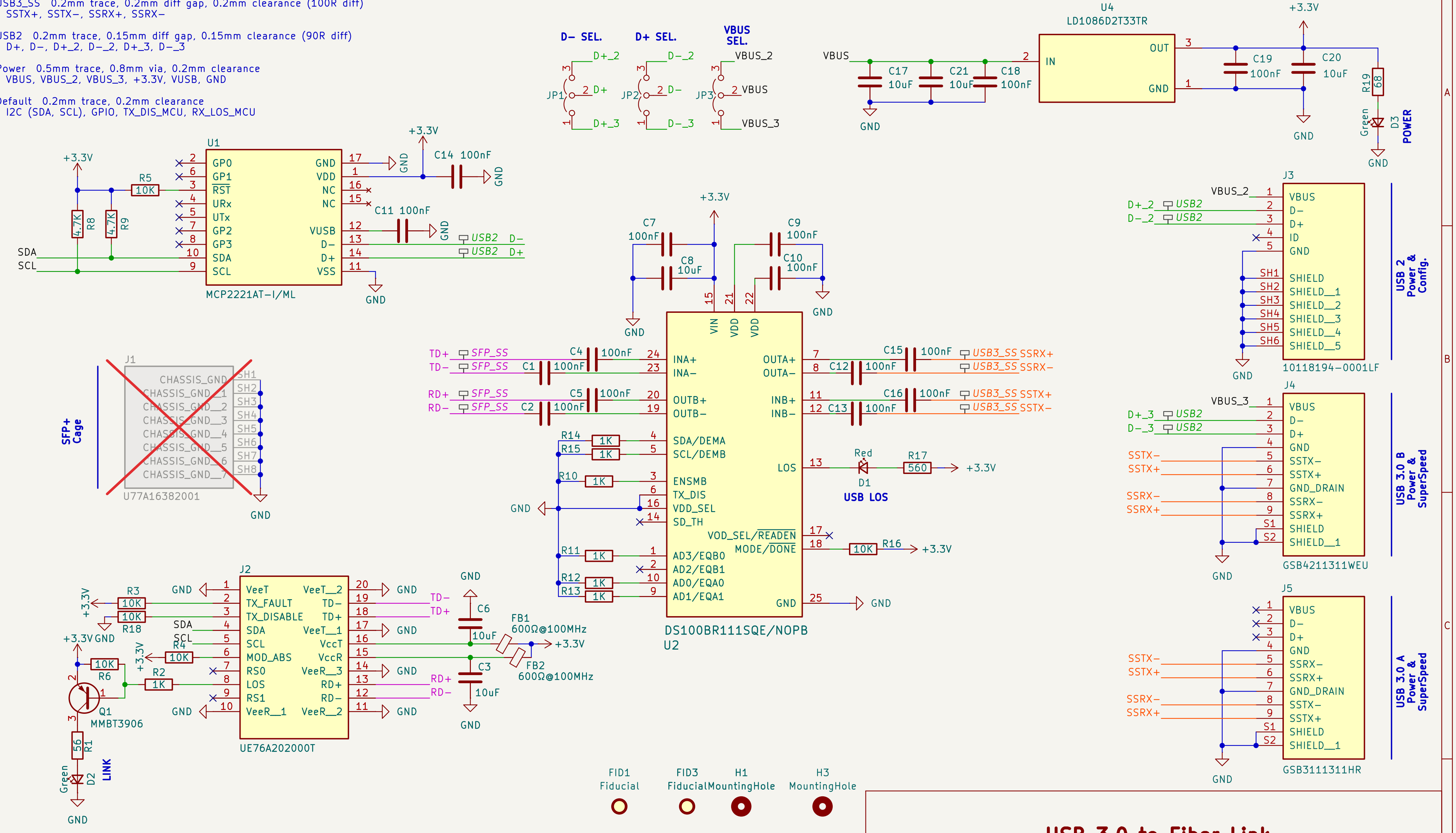
NET CLASSES (via directive label rectangles)

USB3_SS 0.2mm trace, 0.2mm diff gap, 0.2mm clearance (100R diff)
  SSTX+, SSTX−, SSRX+, SSRX−

USB2 0.2mm trace, 0.15mm diff gap, 0.15mm clearance (90R diff)
  D+, D−, D+_2, D−_2, D+_3, D−_3

Power 0.5mm trace, 0.8mm via, 0.2mm clearance
  VBUS, VBUS_2, VBUS_3, +3.3V, VUSB, GND

Default 0.2mm trace, 0.2mm clearance
  I2C (SDA, SCL), GPIO, TX_DIS_MCU, RX_LOS_MCU



NOTES
- EQ: set to minimum/near−minimum for short traces (<25mm). Overdriving EQ on short traces amplifies noise.
- Config resistor values: select from DS100BR111 datasheet table before PCB spin.
- All 100nF bypass caps: 0603 ceramic, place as close to VCC pins as possible.
- All high speed data coupling caps: 0402 ceramic, placed within 2mm of DS100BR111 pins
- LDO output cap: 10uF minimum required for LD1117 stability.
- TX_DISABLE: driven by MCP2221A GP3 (active HIGH disables laser).
- I2C: 100kHz per SFP+ MSA spec. Check clock stretching if module unresponsive.

USB 3.0 to Fiber Link

Repo: github.com/ringof/usb3-fiber

Designer: David Goncalves	Commit: local	OSH Licence: CERN−OHL−P−2.0
Size: A4	Date: 2026−03−13	Rev: DEV
Sheet: /	File: usb3_fiber.kicad_sch	Page 1 of 1